MATHEMATICS / INDEPENDENT STUDY

# MM7033

# Advanced Algebra

Written lectures · eight teaching units

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# Contents

1 Unit 1 · Fields and splitting fields ..... 3
1.1 Lecture 1 · Minimal polynomials and splitting constructions ..... 3
1.2 Lecture 2 · Finite fields and Frobenius ..... 4
2 Unit 2 · Existence and transcendence ..... 6
2.1 Lecture 3 · Zorn's lemma and maximal ideals ..... 6
2.2 Lecture 4 · Algebraic closures and transcendence bases ..... 7
3 Unit 3 · Modules and quotient constructions ..... 9
3.1 Lecture 5 · Modules as linear algebra over a ring ..... 9
3.2 Lecture 6 · Submodules, quotients, and isomorphism theorems ..... 10
4 Unit 4 · Free modules and exact sequences ..... 12
4.1 Lecture 7 · Direct sums, products, and presentations ..... 12
4.2 Lecture 8 · Exactness and splitting ..... 13
5 Unit 5 · Tensor products ..... 15
5.1 Lecture 9 · Linearizing bilinear maps ..... 15
5.2 Lecture 10 · Computation and right exactness ..... 16
6 Unit 6 · Localization ..... 18
6.1 Lecture 11 · Inverting selected scalars ..... 18
6.2 Lecture 12 · Localizing modules and exact sequences ..... 19
7 Unit 7 · Multilinear algebra ..... 21
7.1 Lecture 13 · Exterior powers and alternating maps ..... 21
7.2 Lecture 14 · Intrinsic trace and determinant ..... 22
8 Unit 8 · Noetherian methods and applications ..... 24
8.1 Lecture 15 · Finiteness conditions and Hilbert's basis theorem ..... 24
8.2 Lecture 16 · Algebraic structure in geometry and representations ..... 25

# 1 Unit 1 · Fields and splitting fields

## 1.1 Lecture 1 · Minimal polynomials and splitting constructions

**Learning goals and prerequisites**
Use evaluation kernels, extend embeddings when adjoining roots, and prove uniqueness of splitting fields. Prerequisite: the field-extension part of MM5020.

**Recall and approach.** Recall the quotient description of an algebraic simple extension and the tower law.

A field extension can be studied through equations satisfied by its elements. The crucial distinction is between a polynomial with one chosen root and a field containing all of its roots. Throughout the course, fields and unqualified rings are commutative, rings have an identity, and homomorphisms preserve it.

**Definition 1 · Minimal polynomial and splitting field**
For $\alpha$ algebraic over $K$, the evaluation kernel in $K[x]$ is generated by a unique monic irreducible polynomial $m_{\alpha,K}$. A splitting field of a nonzero $f \in K[x]$ is an extension generated by all roots of $f$ in which $f$ splits into linear factors.

The kernel statement follows by polynomial division: a remainder of smaller degree cannot vanish at $\alpha$. Irreducibility follows because a field has no zero divisors. Consequently $K(\alpha) \cong K[x]/(m)$ and its degree is $\deg m$. A monic irreducible polynomial therefore serves as both a relation and a basis-size certificate.

**Proof strategy.** Define the extension on polynomial expressions and use the minimal-polynomial kernel to justify it.

**Theorem 1 · Extending an embedding**
Let $\sigma : K \to L$ be a field embedding, let $\alpha$ be algebraic over $K$, and suppose $\beta$ in an extension of $L$ is a root of the coefficient-transformed polynomial $\sigma(m_{\alpha,K})$. There is a unique extension $K(\alpha) \to L(\beta)$ sending $\alpha$ to $\beta$.

**Proof.** Send $\sum_i a_i\alpha^i$ to $\sum_i \sigma(a_i)\beta^i$. The transformed minimal polynomial is irreducible over $\sigma(K)$, so its evaluation kernel there is generated by that polynomial. Thus the formula descends through the quotient and is injective. Its values on $K$ and $\alpha$ force uniqueness. The image is $\sigma(K)(\beta)$, which need not be all of $L(\beta)$ unless $\sigma(K) = L$. □

**Theorem 2 · Splitting fields exist and are unique over the base**
Every nonzero polynomial has a finite splitting field (the base field suffices for a constant). Any two splitting fields are isomorphic by an isomorphism fixing the base field.

**Proof.** For existence, adjoin a root of an irreducible factor using its quotient field, divide out a linear factor, and repeat. The process terminates after at most the original degree many root adjunctions, each finite. For uniqueness, extend the identity embedding of the base through a generating list of roots, choosing a root of the transformed minimal polynomial in the second splitting field at each step. Such a root exists because that polynomial divides the remaining transformed polynomial. The final embedding maps the full root set onto the other root set, with multiplicities respected. Since the second field is generated by those roots, the embedding is onto. □

**Example 1 · One root is not always enough**
For $x^3 - 2$ over $\mathbb{Q}$, the real root $\alpha$ generates a real field of degree three. A splitting field also needs a primitive cube root of unity $\zeta$. The field $\mathbb{Q}(\alpha, \zeta)$ has degree six: $\zeta$ is nonreal, so has degree two over the real field $\mathbb{Q}(\alpha)$.

**Example 2 · Embedding into a larger named target**
Send $\mathbb{Q}\left(\sqrt{2}\right)$ into $\mathbb{R}$ by $\sqrt{2} \mapsto -\sqrt{2}$. This is a valid embedding but is not onto $\mathbb{R}$; its image is the same quadratic subfield. The root-extension theorem promises an embedding into the target containing the chosen root. Surjectivity requires a separate generation argument.

In splitting-field uniqueness, that argument succeeds because the image contains every root generating the target. In a general extension problem the target may contain unrelated elements, so the same conclusion would be unjustified. Keep the image field visible throughout the construction.

**Checkpoint 1.1.** Why does a field embedding have trivial kernel?

**Takeaways.** Always state the base field and distinguish an embedding from a surjective isomorphism onto the named target.
Continue with study guide U1 and exercise sheet U1.

## 1.2 Lecture 2 · Finite fields and Frobenius

**Learning goals and prerequisites**
Classify finite fields, identify their subfields, and distinguish repeated roots from irreducibility.

**Recall and approach.** Recall finite-field multiplicative groups are cyclic and every element satisfies its field-size polynomial.

Every finite field has order $p^n$ and is a splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$. Existence follows by taking the roots in a splitting field: Frobenius makes them closed under field operations, and the derivative $-1$ makes the $p^n$ roots distinct. Uniqueness follows from splitting-field uniqueness. We write $\mathbb{F}_{p^n}$ for its isomorphism class.

**Theorem 3 · Subfields of a finite field**
The field $\mathbb{F}_{p^n}$ has a subfield of order $p^d$ exactly when $d|n$, and then that subfield is unique.

**Proof strategy.** Use the tower law for necessity and the fixed points of a Frobenius power for existence.
**Proof.** If a subfield has degree $d$ over $\mathbb{F}_p$, the tower law makes $d|n$. Conversely suppose $d|n$. The nonzero roots of $x^{p^d} - x$ satisfy $x^{p^d-1} = 1$ in the cyclic group of order $p^n - 1$. Since $p^d - 1$ divides $p^n - 1$, there are exactly $p^d - 1$ such roots: in a cyclic group the equation $x^r = 1$ has $r$ solutions when $r$ divides the group order. Together with zero, these form a field by Frobenius closure, of order $p^d$. Any subfield of that size consists of roots of the same polynomial, proving uniqueness. □

**Definition 2 · Separable polynomial**
A polynomial over a field is separable if it has no repeated root in a splitting field. A polynomial has a repeated root exactly when it has a nonconstant common factor with its formal derivative.

The derivative criterion follows by differentiating a factorization into linear factors. At a simple root $a$, the derivative equals the nonzero product of all other root differences times the leading coefficient. At a repeated root, both polynomial and derivative vanish. A common root is equivalent to a common irreducible factor over the base field.

**Theorem 4 · Irreducibles over finite fields are separable**
Every irreducible polynomial over a finite field has distinct roots.

**Proof.** An irreducible with a repeated root must divide its derivative, so the derivative must be zero by degree. In characteristic $p$, this means only exponents divisible by $p$ occur. Frobenius is a bijection on the finite coefficient field, so every coefficient has a $p$th root. The polynomial is therefore a $p$th power of a positive-degree polynomial, contradicting irreducibility. □

**Watch for this.** An irreducible polynomial over an infinite field of characteristic $p$ can be inseparable. In $\mathbb{F}_{p(t)}[x]$, $x^p - t$ is irreducible by Eisenstein at $t$ before passing to the fraction field, but its derivative is zero.

**Example 3 · Subfields form a divisibility pattern**
Inside $\mathbb{F}_{2^{12}}$, the subfields of sizes sixteen and sixty-four have intersection of size four: its degree divides both four and six, and the unique degree-two field lies in both. The field they generate has degree twelve, since its degree is divisible by both four and six and divides twelve.

This reasoning needs uniqueness of subfields inside the ambient finite field, not just existence of abstract fields of those sizes. The intersection and compositum are concrete subfields of one fixed field, so the tower law can compare their degrees.

**Checkpoint 1.2.** Which subfield sizes occur in a field with sixty-four elements?

**Takeaways.** Frobenius simultaneously explains field arithmetic, subfields, and the separability of finite-field polynomials.
Continue with study guide U1 and exercise sheet U1.

# 2 Unit 2 · Existence and transcendence

## 2.1 Lecture 3 · Zorn's lemma and maximal ideals

**Learning goals and prerequisites**
Apply Zorn's lemma with explicit chain upper bounds, prove maximal-ideal existence, and separate algebraic from set-theoretic assumptions.

**Recall and approach.** Recall maximal means no larger admissible object, whereas greatest means above every object.

Some important existence statements cannot be proved by choosing an element of smallest size. Zorn's lemma replaces a size-minimization argument by an order-theoretic condition on chains. We use the usual axiom of choice in this course through the following equivalent formulation.

**Proof strategy.** Specify the poset and verify that a chain union remains admissible before invoking maximality.

**Theorem 5 · Zorn's lemma, foundational assumption**
A nonempty partially ordered set in which every chain has an upper bound has a maximal element. A chain is a subset whose elements are pairwise comparable. A maximal element need not be a greatest element.

The lemma is a set-theoretic principle used here, not a theorem derived from the ring axioms. The mathematical work in an application is to choose the correct poset and verify the upper-bound condition without accidentally including a forbidden object.

**Theorem 6 · Existence of maximal ideals**
Every nonzero commutative unital ring has a maximal ideal. More generally every proper ideal is contained in a maximal ideal.

**Proof.** Fix a proper ideal $I$ and order the proper ideals containing it by inclusion. The poset is nonempty. For a nonempty chain, its union is an ideal: any two elements lie together in one chain member, so their difference lies there, and multiplication by a ring element stays in that member. The union is proper, because if it contained one, one chain member would contain one and would be the whole ring. Thus the union is an upper bound in the poset; the empty chain has upper bound $I$. Zorn supplies a maximal member, which is a maximal ideal of the ring. □

**Example 4 · What the proof does and does not give**
In $\mathbb{Z}$, maximal ideals are $(p)$ for primes $p$. In an arbitrary ring, the argument proves existence without an algorithm for finding a maximal ideal. The requirement $1 \neq 0$ matters: the zero ring has no proper ideal.

**Theorem 7 · Vector-space bases**
Every vector space has a basis, assuming Zorn's lemma.

**Proof.** Order linearly independent subsets by inclusion. Their chain unions are independent because every finite linear dependence would involve elements lying together in one chain member. A maximal independent subset exists. If it did not span, adjoining a vector outside its span would preserve independence, contradicting maximality. □

**Watch for this.** For arbitrary collections of ideals, the union need not be an ideal. Comparability in a chain is what places two selected elements in a common ideal. Similarly, independence is preserved by chain unions because every relation is finite.

**Example 5 · The chain condition does the work**
The ideals $2\mathbb{Z}$ and $3\mathbb{Z}$ are proper, but their union is not an ideal: it contains two and three without containing their sum five. In a chain, one of two selected ideals contains the other, so two chosen elements do lie in a common member and can be added there. Properness uses a different argument: if one belonged to the union, it would belong to a single member.

These are two independent checks. Closure proves the union is an ideal; the identity test proves it is still in the poset of proper ideals. Omitting either leaves a gap in the Zorn argument.

**Checkpoint 2.1.** Where does the identity element enter the maximal-ideal proof?

**Takeaways.** Every Zorn proof needs four visible ingredients: the poset, nonemptiness, chain upper bounds, and the implication of maximality.
Continue with study guide U2 and exercise sheet U2.

## 2.2 Lecture 4 · Algebraic closures and transcendence bases

**Learning goals and prerequisites**
Construct an algebraic closure and prove existence of a transcendence basis. Recall algebraic extensions, polynomial rings, and finite dependence.

**Recall and approach.** Recall an extension generated by finitely many algebraic elements is finite.

**Definition 3 · Algebraic closure**
An algebraic closure of $K$ is an algebraic extension $\frac{\overline{K}}{K}$ in which every nonconstant polynomial has a root. Repeatedly factoring off roots then makes every polynomial split.

**Theorem 8 · Existence of algebraic closures**
Every field has an algebraic closure.

**Proof strategy.** Adjoin roots of all current polynomials at once, then repeat countably many times so that polynomials involving new coefficients also acquire roots. **Proof.** For each monic nonconstant $f \in K[x]$, introduce a variable $X_f$. In the polynomial ring on all these variables, the ideal generated by the elements $f(X_f)$ is proper. Otherwise an expression for one would use finitely many generators; choose a field splitting those finitely many polynomials and evaluate their variables at roots, assigning any other finitely occurring variables arbitrary values. This would send one to zero, impossible. Extend the ideal to a maximal ideal and take the field quotient. The base field embeds because the quotient is nonzero. Every generator image is algebraic, so every element of this field is algebraic over $K$, since it uses finitely many algebraic generators. Call it $K_1$. Repeat to get $K \subseteq K_1 \subseteq K_2 \subseteq \dots$. The union is a field and is algebraic over $K$ by transitivity of algebraicity. A polynomial over the union has finitely many coefficients, all in some $K_n$, and therefore has a root in $K_{n+1}$. The union is an algebraic closure. □

**Definition 4 · Algebraic independence**
A subset $B \subseteq L$ is algebraically independent over $K$ if no nonzero polynomial over $K$ in finitely many distinct elements of $B$ evaluates to zero. A transcendence basis is such a $B$ for which $\frac{L}{K(B)}$ is algebraic.

**Theorem 9 · Existence of transcendence bases**
Every field extension $\frac{L}{K}$ has a transcendence basis.

**Proof.** Order the algebraically independent subsets by inclusion. The empty set qualifies, and a chain union is independent because a polynomial relation uses only finitely many elements, which occur together in one chain member. Zorn gives a maximal independent set $B$. If $\alpha \in L$ were transcendental over $K(B)$, then $B \cup \{\alpha\}$ would be independent: a polynomial relation, viewed as a polynomial in $\alpha$, has coefficients that are polynomials in finitely many elements of $B$, not all zero by independence. It would make $\alpha$ algebraic over $K(B)$, a contradiction. Thus $\frac{L}{K(B)}$ is algebraic. □

**Example 6 · Rational functions**
$\{t_1, \ldots, t_n\}$ is a transcendence basis of $K(t_1, \ldots, t_n)$ over $K$. Adjoining $\sqrt{t_1}$ in characteristic not two gives an algebraic extension without changing that chosen transcendence basis. A transcendence basis need not generate the whole field as a field.

**Example 7 · A transcendence basis need not generate the field**
In $K(t)$, the singleton $\{t^2\}$ is algebraically independent, since a nonzero polynomial in $t^2$ would produce a nonzero polynomial in the indeterminate $t$. Yet $t$ satisfies $X^2 - t^2$ over $K(t^2)$, so $K(t)$ is algebraic over that subfield. Thus $\{t^2\}$ is a transcendence basis even though adjoining it as a rational-function generator need not recover all of $K(t)$.

The definition has two clauses: independence over the original base and algebraicity over the generated field. Maximality supplies the second only after proving that an element transcendental over that field could be adjoined without destroying independence.

**Checkpoint 2.2.** Why is adjoining roots only once insufficient for an algebraic closure?

**Takeaways.** The common mechanism is finite dependence inside a maximal or increasing construction. Keep it visible in both existence proofs.
Continue with study guide U2 and exercise sheet U2.

# 3 Unit 3 · Modules and quotient constructions

## 3.1 Lecture 5 · Modules as linear algebra over a ring

**Learning goals and prerequisites**
Verify module axioms, distinguish generation from freeness, and translate examples into module language.

**Recall and approach.** Recall an abelian group already has its unique integer-scalar action by repeated addition.

A vector space allows division by every nonzero scalar. A module keeps addition and scalar multiplication while dropping that assumption. Consequently bases may fail to exist, submodules of free modules need not be direct summands, and a nonzero scalar can annihilate a nonzero vector.

**Definition 5 · Module**
For a commutative unital ring $R$, an $R$-module is an abelian group $M$ with a scalar action satisfying $r(m+n) = rm + rn$, $(r+s)m = rm + sm$, $(rs)m = r(sm)$, and $1m = m$. A module map is an additive map satisfying $f(rm) = rf(m)$.

Every abelian group is a $\mathbb{Z}$-module by repeated addition, and every $\mathbb{Z}$-module arises this way. The quotient $\frac{R}{I}$ is an $R$-module. An ideal is a submodule of the regular module $R$. A vector space over $K$ with a linear operator $T$ becomes a $K[x]$-module by $f(x) \cdot v = f(T)v$. Conversely the action of $x$ determines such an operator, and module maps are exactly linear maps commuting with it.

**Definition 6 · Generation and annihilator**
Elements $m_i$ generate $M$ if every element is a finite sum $\sum_i r_i m_i$. The annihilator of $M$ is $\mathrm{Ann}_{R(M)} = \{r : rM = 0\}$, an ideal. For one element, $\mathrm{Ann}_{R(m)}$ is the kernel of $R \to M$, $r \mapsto rm$.

**Proof strategy.** Map the regular module onto a cyclic module and identify its annihilator; use basis values to control maps from free modules.

**Theorem 10 · Cyclic modules**
A module generated by one element $m$ is isomorphic to $R/\mathrm{Ann}_{R(m)}$.

**Proof.** The map $R \to M$ sending $r$ to $rm$ is linear and onto by cyclicity. Its kernel is the displayed annihilator. The quotient map sends $[r]$ to $rm$; equality of cosets is exactly equality of these values, so the induced map is a bijective module map. □

**Example 8 · One generator need not be a basis**
The $\mathbb{Z}$-module $\mathbb{Z}/6\mathbb{Z}$ is generated by $[1]$, but $6[1] = 0$ is a nontrivial scalar relation. It is not free: a nonzero free $\mathbb{Z}$-module has elements whose nonzero integer multiples never vanish.

**Theorem 11 · Module maps from a free module**
A map from $R^n$ to $M$ is uniquely determined by arbitrary images of the standard basis vectors.

**Proof.** The only possible formula is $(r_1, \ldots, r_n) \mapsto \sum_i r_i m_i$. This formula is linear and has the desired basis values, proving both existence and uniqueness. □

**Example 9 · An operator expressed as a module relation**
Let $T$ on $K^2$ satisfy $T(e_1) = 0$ and $T(e_2) = e_1$. As a $K[x]$-module, $e_2$ generates the entire space: applying $x$ gives $e_1$. Applying $x^2$ gives zero, and a polynomial kills $e_2$ exactly when its constant and linear coefficients vanish. The module is therefore $\frac{K[x]}{x^2}$.

Its dimension as a $K$-vector space is two, but it has one generator as a $K[x]$-module and is not free over $K[x]$. The coefficient ring determines what "generator," "basis," and "dimension" are allowed to mean.

**Checkpoint 3.1.** What is the annihilator of $\mathbb{Z}/12\mathbb{Z}$ as a $\mathbb{Z}$-module?

**Takeaways.** Ask separately whether a module is generated by some elements and whether the expression in those generators is unique.
Continue with study guide U3 and exercise sheet U3.

## 3.2 Lecture 6 · Submodules, quotients, and isomorphism theorems

**Learning goals and prerequisites**
Construct quotient modules, identify kernels and images, and use the correspondence and isomorphism theorems.

**Recall and approach.** Recall well-defined scalar action on additive cosets follows from submodule closure.

**Definition 7 · Submodule and quotient**
A submodule $N \subseteq M$ is an additive subgroup closed under all scalars. The quotient $M/N$ consists of additive cosets, with $r(m + N) = rm + N$. A kernel and an image of module maps are submodules.

The scalar action on the quotient is well defined because if $m - m' \in N$, then $r(m - m') \in N$. This is the module analogue of normality in group quotients, but additive groups of modules are already abelian.

**Proof strategy.** Restrict or compose quotient maps and compute their kernels; use correspondence for simple modules.

**Theorem 12 · First module isomorphism theorem**
For a linear map $f : M \to P$, the induced map $M/\ker f \to \operatorname{im} f$, $m + \ker f \mapsto f(m)$, is an isomorphism.

**Proof.** The additive-group theorem gives a well-defined bijective additive map. Scalar compatibility follows from $f(rm) = rf(m)$; its inverse is linear because the map is bijective and preserves scalar operations. □

**Theorem 13 · Second and third module isomorphism theorems**
For submodules $A, B$ of $M$, $(A + B)/B \cong A/(A \cap B)$. For $A \subseteq B \subseteq M$, $(M/A)/(B/A) \cong M/B$.

**Proof.** Restrict the projection $M \to M/B$ to $A$. Its image is $(A + B)/B$ and its kernel is $A \cap B$, proving the first result. For the second, the map $m + A \mapsto m + B$ is well defined because $A \subseteq B$, is surjective, and has kernel $B/A$. Apply the first isomorphism theorem. □

Submodules of $M/N$ correspond to submodules of $M$ containing $N$, by inverse image under the quotient map. Image followed by inverse image returns a submodule containing $N$; the reverse composite returns the original quotient submodule because the quotient map is onto. This correspondence is useful for recognizing maximal submodules and simple modules.

**Definition 8 · Simple module**
A nonzero module is simple if its only submodules are zero and itself.

**Theorem 14 · Simple modules over a commutative ring**
A simple $R$-module is isomorphic to $R/\mathfrak{m}$ for a maximal ideal $\mathfrak{m}$, and every such quotient is simple.

**Proof.** A nonzero element generates a nonzero submodule, hence all of a simple module. The cyclic-module theorem identifies it with $\frac{R}{I}$. Submodule correspondence says this quotient is simple exactly when no ideal lies strictly between $I$ and $R$, namely when $I$ is maximal. □

**Example 10 · An intersection calculation behind an isomorphism**
In the integer module, $4\mathbb{Z} + 6\mathbb{Z} = 2\mathbb{Z}$ by Bézout and $4\mathbb{Z} \cap 6\mathbb{Z} = 12\mathbb{Z}$ by divisibility. The map $4\mathbb{Z} \to 2\frac{\mathbb{Z}}{6}\mathbb{Z}$ has kernel $12\mathbb{Z}$ and is onto: $[4]$ generates the three-element target. It gives $4\frac{\mathbb{Z}}{12}\mathbb{Z} \cong 2\frac{\mathbb{Z}}{6}\mathbb{Z}$.

This concrete example explains why the second isomorphism theorem has an intersection on one side and a sum on the other. The domain restriction creates the intersection kernel, while taking its image in the quotient creates the sum modulo the other submodule. Memorizing the formula without this map makes its direction easy to reverse.

**Checkpoint 3.2.** Why is $\mathbb{Z}/9\mathbb{Z}$ not a simple $\mathbb{Z}$-module?

**Takeaways.** Many module proofs reduce to choosing a natural linear map and computing its kernel and image.
Continue with study guide U3 and exercise sheet U3.

# 4 Unit 4 · Free modules and exact sequences

## 4.1 Lecture 7 · Direct sums, products, and presentations

**Learning goals and prerequisites**
Distinguish sums from products, use free-module universal properties, and interpret finite presentations.

**Recall and approach.** Recall finite-support tuples are the algebraic direct sum, while products allow arbitrary support.

**Definition 9 · Sum and product**
The product $\prod_i M_i$ consists of all coordinate tuples. The direct sum $\bigoplus_i M_i$ consists of tuples with finite support. Both have coordinatewise operations; they coincide for finite families.

The product is characterized by maps into its factors: a family $X \to M_i$ uniquely assembles into $X \to \prod_i M_i$. The direct sum is characterized by maps out of its summands: maps $M_i \to X$ assemble by a finite sum into $\bigoplus_i M_i \to X$. Finite support is essential because arbitrary infinite algebraic sums are not defined.

**Definition 10 · Free module and presentation**
A free module on a set $S$ is $R^S$, the finite-support functions $S \to R$, with standard basis $e_s$. A presentation of $M$ is an exact sequence $R^T \to R^S \to M \to 0$. The elements of $S$ name generators; images of the basis elements indexed by $T$ name relations.

**Proof strategy.** Build a free cover from chosen generators and then cover its kernel to record relations.

**Theorem 15 · Every module has a presentation**
Every $R$-module is a quotient of a free module and admits a presentation.

**Proof.** Take the free module on the underlying set of $M$ and send the basis vector $[m]$ to $m$. This is surjective. Apply the same construction to its kernel and compose that free-module map with the kernel inclusion. The image of the first map is exactly the kernel of the second, which is exactness. □

A module is finitely generated when the generator set can be finite. It is finitely presented when both generator and relation sets can be finite. Over a Noetherian ring these coincide for finitely generated modules, as Unit 8 will show; over an arbitrary ring they need not coincide.

**Example 11 · A finite presentation**
$\mathbb{Z}/6\mathbb{Z}$ has presentation $\mathbb{Z} \xrightarrow{\times 6} \mathbb{Z} \to \mathbb{Z}/6\mathbb{Z} \to 0$. The single relation is $6e = 0$. Over $R = K[x]$, the module $R/(x^2)$ has basis $1, x$ as a $K$-vector space, and multiplication by $x$ is nilpotent of order two.

**Theorem 16 · Finite free rank is well defined**
If $R$ is a nonzero commutative ring and $R^m \cong R^n$ for finite $m, n$, then $m = n$.

**Proof.** Choose a maximal ideal $\mathfrak{m}$. The isomorphism induces an isomorphism of quotients by $\mathfrak{m}$ times each module, giving $(R/\mathfrak{m})^m \cong (R/\mathfrak{m})^n$. Vector-space dimension over the residue field forces equality.□

**Example 12 · Reading a presentation without guessing a basis**
A module with generators $u, v$ and relations $u - 2v = 0$, $6v = 0$ is isomorphic to $\frac{\mathbb{Z}}{6}\mathbb{Z}$: send $v$ to one modulo six and $u$ to two. These values respect both relations. The reverse map sends one to $v$ and is well defined because $6v = 0$. The composites fix the generators, hence every element.

Eliminating a generator from a relation suggests the answer, but the two maps prove no additional relation has been introduced or lost. This distinction matters when several relations interact and an informal simplification may accidentally discard information.

**Checkpoint 4.1.** What fails if the ring in the rank theorem is the zero ring?

**Takeaways.** Generators and relations describe a module without asserting it has a basis. This language will make tensor computations systematic.
Continue with study guide U4 and exercise sheet U4.

## 4.2 Lecture 8 · Exactness and splitting

**Learning goals and prerequisites**
Read exact sequences, prove the splitting lemma, and identify the lifting property of free and projective modules.

**Recall and approach.** Recall exactness identifies an image with a kernel; it does not by itself supply a complementary summand.

**Definition 11 · Exact sequence**
A sequence $A \xrightarrow{i} B \xrightarrow{p} C$ is exact at $B$ if $\operatorname{im} i = \ker p$. A short exact sequence is $0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$, so $i$ is injective, $p$ is surjective, and their image/kernel coincide.

**Proof strategy.** Use a section to decompose each middle element, and use projective lifting to construct such sections.

**Theorem 17 · Splitting lemma**
For a short exact sequence, the following are equivalent: $p$ has a linear section $s : C \to B$; $i$ has a linear retraction $r : B \to A$; and $B$ is isomorphic to $A \oplus C$ compatibly with the given maps.

**Proof.** Given a section, every $b$ decomposes uniquely as $i(a) + s(p(b))$: the difference $b - s(p(b))$ lies in $\ker p = \operatorname{im} i$, and injectivity determines $a$. The map $(a, c) \mapsto i(a) + s(c)$ is therefore an isomorphism, giving a retraction by taking the first coordinate. Conversely, given a retraction $r$, the restriction of $p$ to $\ker r$ is bijective onto $C$. It is injective because an element in both $\ker r$ and $\ker p$ is $i(a)$ with $a = ri(a) = 0$. It is onto because replacing a lift $b$ of $c$ by $b - i(r(b))$ puts it in $\ker r$. Its inverse followed by inclusion is a section. The direct-sum form visibly gives both maps. □

**Definition 12 · Projective module**
A module $P$ is projective if for every surjection $q : M \to N$ and map $f : P \to N$, there is a lift $h : P \to M$ with $qh = f$.

**Theorem 18 · Free and projective modules**
Free modules are projective. A module is projective exactly when it is a direct summand of a free module.

**Proof.** For a free module, choose a preimage under $q$ for the image of each basis vector and extend linearly. A direct summand inherits this lifting property by extending a map through the projection from the free module, lifting, and restricting to the summand. Conversely, a free module surjects onto any $P$. If $P$ is projective, lift its identity to obtain a section of that surjection. The splitting lemma realizes $P$ as a direct summand. □

**Example 13 · A nonsplit sequence**
$0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ is exact but does not split. A section would send $[1]$ to an integer annihilated by two, necessarily zero, which cannot map back to $[1]$.

**Example 14 · Finite projective means a summand of finite free**
If a projective module $P$ is generated by $p_1, \ldots, p_n$, the corresponding map $R^n \to P$ is onto. Projectivity lifts $1_P$ to a section $P \to R^n$, and the splitting lemma makes $P$ a direct summand of this finite free module. This justifies the finite dual-basis construction used for trace later.

For $R = K \times K$, the module $R(1,0)$ is such a summand, complementary to $R(0,1)$. It is projective but not free: the nonzero scalar $(0,1)$ annihilates every element, whereas no nonzero free module is annihilated by a nonzero scalar. Thus a splitting is weaker than a basis.

**Checkpoint 4.2.** Why does every short exact sequence ending in a free module split?

**Takeaways.** Exactness records what is lost and retained. Splitting is the stronger assertion that the two pieces can be independently recovered inside the middle module.
Continue with study guide U4 and exercise sheet U4.

# 5 Unit 5 · Tensor products

## 5.1 Lecture 9 · Linearizing bilinear maps

**Learning goals and prerequisites**
Construct a tensor product, use its universal property, and distinguish pure tensors from arbitrary tensors. All modules in this unit are over a commutative ring $R$.

**Recall and approach.** Recall a map out of a quotient must kill its defining relations.

A bilinear map accepts two module elements and is linear in each variable. The tensor product replaces this two-input mapping problem by an ordinary linear mapping problem. Its elements are formal combinations constrained by bilinearity, not pairs with componentwise operations.

**Definition 13 · Tensor product**
The tensor product $M \otimes_R N$ is a module with a bilinear map $(m, n) \mapsto m \otimes n$ such that every bilinear $b : M \times N \to P$ factors uniquely as a linear map $\overline{b} : M \otimes_R N \to P$ satisfying $\overline{b}(m \otimes n) = b(m, n)$.

**Proof strategy.** Begin with a free module on symbols and impose exactly the bilinearity relations; maps descend precisely when those relations vanish.

**Theorem 19 · Existence by generators and relations**
Start with the free $R$-module on symbols $[m, n]$. Quotient by the submodule generated by

$$[m + m', n] - [m, n] - [m', n], \quad [m, n + n'] - [m, n] - [m, n'], \tag{1}$$

$$[rm, n] - r[m, n], \quad [m, rn] - r[m, n]. \tag{2}$$

The quotient, with $m \otimes n$ the class of $[m, n]$, is a tensor product.

**Proof.** The relations make the symbol map bilinear. A bilinear $b$ extends uniquely from the free basis to a linear map on the free module, and the displayed relations lie in its kernel. It therefore descends uniquely to the quotient. Conversely any linear map from the quotient gives a bilinear map by composition. □

Every tensor is a finite sum of pure tensors $m \otimes n$. It need not itself be pure. For vector spaces, $e_1 \otimes e_1 + e_2 \otimes e_2$ in $K^2 \otimes K^2$ is not pure: a pure tensor gives a coefficient matrix of rank at most one, while this tensor gives the identity matrix of rank two.

**Theorem 20 · Basic tensor identifications**
There are canonical isomorphisms $R \otimes_R M \cong M$, $M \otimes_R N \cong N \otimes_R M$, and $(M \otimes_R N) \otimes_R P \cong M \otimes_{R(N \otimes_R P)}$.

**Proof.** For the first, send $r \otimes m$ to $rm$, with inverse $m \mapsto 1 \otimes m$. The tensor relations verify both composites on pure tensors. For symmetry, interchange the two inputs; the operation is bilinear and is its own inverse. For associativity, both objects classify trilinear maps from $M \times N \times P$; equivalently define the map on pure tensors by $(m \otimes n) \otimes p \mapsto m \otimes (n \otimes p)$, check multilinearity through the two universal properties, and reverse the parentheses for its inverse. Pure tensors generate, so the inverse equations hold everywhere. □

**Example 15 · Why a tempting tensor formula fails**
Over $\mathbb{Z}$, the proposal $m \otimes n \mapsto m + n$ from $\mathbb{Z} \otimes \mathbb{Z}$ to $\mathbb{Z}$ is not well defined by the universal property. The proposed two-input function is not bilinear: at $(0, n)$ it gives $n$, whereas a bilinear function must vanish when either input is zero. In contrast $m \otimes n \mapsto mn$ is induced by multiplication and identifies the tensor with $\mathbb{Z}$.

Testing zeros and scalar movement is a fast preliminary check, but a valid proof checks additivity and scalar compatibility in each argument. Only then may a formula on pure tensors be extended linearly.

**Checkpoint 5.1.** Why is $rm \otimes n = m \otimes rn$?

**Takeaways.** Define maps on pure tensors only after verifying the relevant bilinearity relations. A formula on symbols is not automatically a well-defined linear map.
Continue with study guide U5 and exercise sheet U5.

## 5.2 Lecture 10 · Computation and right exactness

**Learning goals and prerequisites**
Compute tensors with quotients, prove right exactness, and exhibit the failure of left exactness.

**Recall and approach.** Recall that pure tensors generate, though not every tensor is pure.

**Proof strategy.** Translate a quotient presentation into tensor relations, then identify the resulting kernel as an image.

**Theorem 21 · Tensoring with a quotient ring**
For an ideal $I$, $(R/I) \otimes_R M \cong M/IM$, through $[r] \otimes m \mapsto rm + IM$.

**Proof.** Changing $r$ by an element of $I$ changes $rm$ by an element of $IM$, so the formula is well defined and bilinear. Its inverse sends $m + IM$ to $[1] \otimes m$. Every generator im of $IM$ maps to $[i] \otimes m = 0$, so this inverse is well defined. The composites are identities on the indicated generators. □

**Example 16 · Tensor products of finite cyclic groups**
Over $\mathbb{Z}$, $(\mathbb{Z}/m\mathbb{Z}) \otimes (\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/\gcd(m, n)\mathbb{Z}$. Apply the quotient formula to obtain $\frac{\mathbb{Z}/n\mathbb{Z}}{m(\mathbb{Z}/n\mathbb{Z})}$, which is $\mathbb{Z}/(m\mathbb{Z} + n\mathbb{Z})$. Bézout identifies the sum ideal with $\gcd(m, n)\mathbb{Z}$.

**Theorem 22 · Right exactness**
If $A \xrightarrow{f} B \xrightarrow{g} C \to 0$ is exact, then $A \otimes N \to B \otimes N \to C \otimes N \to 0$ is exact.

**Proof strategy.** Identify $C$ as the quotient of $B$ by the image of $f$, then show tensoring imposes exactly the corresponding generated relations. **Proof.** Write $C = \frac{B}{K}$ with $K = \operatorname{im} f$. The quotient of $B \otimes N$ by the submodule generated by $k \otimes n$, $k \in K$, has the tensor universal property for $\left(\frac{B}{K}\right) \otimes N$: a bilinear map on $B \times N$ descends to $\left(\frac{B}{K}\right) \times N$ exactly when it vanishes on $K \times N$. Thus it is canonically $C \otimes N$. Its kernel is precisely the submodule generated by the $f(a) \otimes n$, which is the image of $f \otimes 1$. Surjectivity follows because every pure tensor $c \otimes n$ has a lift $b \otimes n$. □

**Example 17 · Injectivity can be destroyed**
Tensor the injection $\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$ with $\mathbb{Z}/2\mathbb{Z}$. Under the identifications with $\mathbb{Z}/2\mathbb{Z}$, the resulting map is multiplication by two, hence zero. Its source is nonzero, so it is not injective.

**Definition 14 · Flatness**
A module $N$ is flat if tensoring with it preserves injections; together with right exactness this means it preserves short exact sequences. Free modules are flat because tensoring with a free module gives a direct sum of copies, and direct sums preserve injections coordinatewise.

**Example 18 · Computing a tensor in two complementary ways**
For $\frac{\mathbb{Z}}{6}\mathbb{Z} \otimes \frac{\mathbb{Z}}{10}\mathbb{Z}$, the generator $[1] \otimes [1]$ is killed by both six and ten, hence by their gcd two. The quotient formula identifies the whole tensor with $\frac{\mathbb{Z}}{2}\mathbb{Z}$, proving the generator has exactly that order rather than merely order dividing two. Equivalently, multiplication modulo two gives a bilinear map detecting it as nonzero.

An annihilation calculation gives an upper bound on size; a detecting map often supplies the missing lower bound. This pattern is useful whenever a formal relation seems to make a tensor vanish but no explicit inverse or universal property has yet been checked.

**Checkpoint 5.2.** What is $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$ for positive $n$?

**Takeaways.** Use presentations to calculate tensors. Preserve surjectivity and the image/kernel relation, but never assume a tensor map remains injective without a flatness argument.
Continue with study guide U5 and exercise sheet U5.

# 6 Unit 6 · Localization

## 6.1 Lecture 11 · Inverting selected scalars

**Learning goals and prerequisites**
Construct localization even in the presence of zero divisors and prove its universal property.

**Recall and approach.** Recall a zero divisor can become zero when another element is made invertible.

Fraction fields invert every nonzero element of a domain. Localization inverts a chosen multiplicatively closed subset and works even when the original ring has zero divisors. We allow a localization to be the zero ring; this occurs, in particular, when the chosen subset contains zero.

**Definition 15 · Localization**
Let $S \subseteq R$ contain one and be closed under multiplication. On pairs $(a, s)$, define $(a, s) \sim (b, t)$ if some $u \in S$ satisfies $u(ta - sb) = 0$. The class is denoted $\frac{a}{s}$. Addition and multiplication use the usual fraction formulas. The resulting ring is $S^{-1}R$.

The extra factor $u$ is essential. It records that an equality becomes true after inverting elements of $S$, even if ordinary cancellation was invalid in $R$. For transitivity, if $u(ta - sb) = 0$ and $v(rb - tc) = 0$, then $uvt(ra - sc) = 0$, so the relation is transitive. Reflexivity and symmetry are immediate. Multiplying cross-differences by suitable denominators verifies that the operations descend to classes.

**Proof strategy.** Use the denominator-annihilator relation rather than cancellation, then force the extension formula by inverses.

**Theorem 23 · Universal property**
A ring map $f : R \to T$ sending every element of $S$ to a unit extends uniquely to $S^{-1}R \to T$, through $\frac{a}{s} \mapsto f(a)f(s)^{-1}$.

**Proof.** If $u(ta - sb) = 0$, then applying $f$ and cancelling the unit images of $u, s, t$ gives equality of the proposed values. The fraction formulas show this is a unital ring homomorphism. Any extension must send the inverse of $\frac{s}{1}$ to the inverse of $f(s)$, forcing the formula and uniqueness. □

**Theorem 24 · Vanishing criterion**
The element $\frac{a}{s}$ is zero exactly when some $u \in S$ satisfies $ua = 0$.

**Proof.** Compare $(a, s)$ with $(0, 1)$ in the defining relation. The condition is exactly $u(a - 0) = 0$. □

**Example 19 · A localization that kills elements**
In $R = \mathbb{Z}/6\mathbb{Z}$, invert the powers of two. The element $[3]$ becomes zero because $[2][3] = 0$. The localization is isomorphic to $\mathbb{F}_3$: reduction modulo three sends two to a unit, and every localized fraction has a unique value there. Inverting nonunits can identify formerly distinct elements.

For a prime ideal $\mathfrak{p}$, the complement $R \setminus \mathfrak{p}$ is multiplicatively closed. The localization is written $R_{\mathfrak{p}}$. If $R$ is a domain and $S = R \setminus \{0\}$, the construction is its fraction field. If $S = \{1, f, f^2, \ldots\}$, write $R_f$.

**Example 20 · Cross multiplication alone is too strict**
In $\frac{\mathbb{Z}}{6}\mathbb{Z}$ localized at powers of two, the fractions $\frac{3}{1}$ and $\frac{0}{1}$ are equal, although their naive cross-difference is the nonzero class three. Multiplication by the allowed denominator two kills that difference. This example is the reason for the extra factor in the equivalence relation.

If zero lies in the denominator set, the vanishing criterion kills every numerator and the localization is the zero ring. Under the convention allowing the zero ring, this causes no contradiction: any target in which zero becomes a unit must itself be the zero ring. State that convention when discussing such degenerate cases.

**Checkpoint 6.1.** Why can the complement of a nonprime ideal fail to be multiplicatively closed?

**Takeaways.** In a ring with zero divisors, equality of localized fractions is not just cross multiplication. Use the explicit vanishing criterion.
Continue with study guide U6 and exercise sheet U6.

## 6.2 Lecture 12 · Localizing modules and exact sequences

**Learning goals and prerequisites**
Construct localized modules, prove localization is exact, and identify localization with tensoring.

**Recall and approach.** Recall the vanishing criterion and how tensoring with a localized ring reexpresses fractions of module elements.

For an $R$-module $M$, define $S^{-1}M$ using pairs $(m, s)$ and the relation $(m, s) \sim (n, t)$ when some $u \in S$ kills $tm - sn$. Scalar multiplication by $\frac{a}{r} \in S^{-1}R$ is $\left(\frac{a}{r}\right)\left(\frac{m}{s}\right) = a\frac{m}{rs}$. The same checks as for rings prove the construction is well defined.

**Proof strategy.** Clear a localized kernel condition with one denominator factor and then use original exactness to lift a numerator.

**Theorem 25 · Exactness of localization**
Localization takes every exact sequence of modules to an exact sequence.

**Proof.** Let $A \xrightarrow{f} B \xrightarrow{g} C$ be exact at $B$. Certainly the localized composite is zero. If $\frac{g(b)}{s} = 0$, the vanishing criterion gives $u \in S$ with $ug(b) = 0$. Thus $ub \in \ker g = \mathrm{im} f$, so $ub = f(a)$ for some $a$. Then $\frac{b}{s} = \frac{f(a)}{us}$, showing the localized kernel lies in the localized image. If $f$ was injective, $\frac{f(a)}{s} = 0$ gives $uf(a) = 0$, hence $ua = 0$, so $\frac{a}{s} = 0$; injectivity is preserved. Surjectivity is preserved by lifting a numerator. □

**Theorem 26 · Localization as tensor product**
There is a canonical isomorphism $S^{-1}R \otimes_R M \cong S^{-1}M$, sending $\left(\frac{a}{s}\right) \otimes m$ to $a\frac{m}{s}$. In particular $S^{-1}R$ is a flat $R$-module.

**Proof.** The displayed formula is bilinear and respects fraction equality. The inverse sends $\frac{m}{s}$ to $\left(\frac{1}{s}\right) \otimes m$. To see it is well defined, if $u(tm - sn) = 0$, then the difference of the proposed images equals $\left(\frac{1}{ust}\right) \otimes u(tm - sn) = 0$, using scalar movement. Composites on pure tensors and fractions are identities. Exactness of localization then gives flatness. □

**Theorem 27 · Localization at a prime is local**
The ring $R_{\mathfrak{p}}$ has a unique maximal ideal, consisting of fractions with numerator in $\mathfrak{p}$.

**Proof.** A numerator outside $\mathfrak{p}$ is in the denominator set, so its fraction is a unit. Fractions with numerator in $\mathfrak{p}$ form an ideal. It is proper: equality $\frac{a}{s} = 1$ with $a \in \mathfrak{p}$ would give $u(a - s) = 0$ for some $u \notin \mathfrak{p}$, hence $us = ua \in \mathfrak{p}$, contradicting primeness. Every element outside this ideal is a unit, so the ideal is maximal and every proper ideal lies in it. □

**Example 21 · Local information at a prime**
$\mathbb{Z}_{(p)}$ consists of rational numbers whose denominators are not divisible by $p$. Its maximal ideal consists of those with numerator divisible by $p$. Integers coprime to $p$ become units; $p$ remains a nonunit.

**Example 22 · Localizing torsion at different primes**
Consider the integer module $\frac{\mathbb{Z}}{12}\mathbb{Z}$. At the prime two, three is invertible, so its localization is the cyclic module killed by four, naturally $\frac{\mathbb{Z}}{4}\mathbb{Z}$ with the induced localized scalar action. At the prime three, four is invertible and the result is $\frac{\mathbb{Z}}{3}\mathbb{Z}$. At a prime dividing neither twelve, twelve itself is invertible and the module becomes zero.

This calculation makes "local information" concrete: different localizations retain different prime parts. It also explains why a single localization need not detect every nonzero module, whereas testing at all maximal ideals does, as proved on the exercise sheet.

**Checkpoint 6.2.** If $S^{-1}M = 0$ and $M$ is finitely generated, what single-annihilator conclusion follows?

**Takeaways.** Localization preserves exact algebraic relationships. The numerator test makes this preservation concrete and is often the shortest route through a calculation.
Continue with study guide U6 and exercise sheet U6.

# 7 Unit 7 · Multilinear algebra

## 7.1 Lecture 13 · Exterior powers and alternating maps

**Learning goals and prerequisites**
Construct exterior powers through a universal property and derive their basis in the free case.

**Recall and approach.** Recall alternating means vanishing on repeated inputs, including in characteristic two.

A multilinear map is alternating if it vanishes whenever two inputs are equal. This definition works in every characteristic. It implies that interchanging two inputs changes the sign: expand the zero value with those two slots both equal to $x + y$. In characteristic two, the sign disappears, but the vanishing condition remains stronger than a bare symmetry statement.

**Definition 16 · Exterior power**
The module $\wedge^n M$ is the quotient of the $n$-fold tensor product by the submodule generated by pure tensors having equal inputs in two positions. The class of $m_1 \otimes \ldots \otimes m_n$ is written $m_1 \wedge \ldots \wedge m_n$. Set $\wedge^0 M = R$.

**Proof strategy.** Use multilinearity for spanning and alternating coordinate detectors for independence.

**Theorem 28 · Exterior universal property**
Alternating $n$-linear maps $M^n \to P$ correspond uniquely to linear maps $\wedge^n M \to P$.

**Proof.** The repeated tensor universal property first corresponds to multilinear maps on $M^n$. Such a map descends through the indicated quotient exactly when it vanishes on the generators with repeated inputs, which is the alternating condition. The quotient factorization is unique. □

**Theorem 29 · Exterior basis**
If $M$ is free with ordered basis $e_1, \ldots, e_r$, then $\wedge^n M$ is free with basis $e_{i_1} \wedge \ldots \wedge e_{i_n}$ for $i_1 < \ldots < i_n$. It is zero for $n > r$ and has rank $\binom{r}{n}$ otherwise.

**Proof strategy.** Expand to prove spanning, then construct coordinate detectors to prove independence.
**Proof.** Multilinearity expands any wedge into basis wedges. Repeated indices give zero, and interchanging adjacent inputs lets us sort the remaining indices with the appropriate sign, proving spanning. For each increasing index set $I$, define the alternating form that takes the determinant of the matrix of the $I$-coordinates of the inputs. It induces a linear functional on the exterior power. On a sorted basis wedge it is one for the matching index set and zero for every other set. These coordinate functionals prove independence over any commutative ring. □

**Example 23 · Area as an algebraic coefficient**
In $R^2$, $(ae_1 + be_2) \wedge (ce_1 + de_2) = (ad - bc)e_1 \wedge e_2$. This is an identity of modules; when $R = \mathbb{R}$, the coefficient also measures signed area. The algebraic definition does not require a metric.

An endomorphism $T : M \to M$ induces $\wedge^n T$ by applying $T$ to each input. Alternation ensures this descends, and the formula immediately gives functoriality: exterior powers preserve identity and composition.

**Example 24 · An exterior coefficient computed twice**
In $R^3$, take $u = e_1 + e_2$ and $v = e_2 + e_3$. Expansion gives $u \wedge v = e_1 \wedge e_2 + e_1 \wedge e_3 + e_2 \wedge e_3$. The three coefficients also equal the two-by-two minors formed from the relevant pairs of coordinates of $u, v$. These coordinate forms detect the three independent basis wedges.

Over a field, $u \wedge v = 0$ exactly when the two vectors are dependent: if $u$ is nonzero, extend it to a basis and inspect the coefficients of $u \wedge v$. Over a ring, dependence and rank require more care, so this field conclusion should not be transferred without hypotheses.

**Checkpoint 7.1.** Why does $e_1 \wedge e_2 + e_2 \wedge e_1 = 0$ hold without dividing by two?

**Takeaways.** The exterior construction is determined by alternating maps. Coordinate determinants prove the basis theorem rather than serving as an unexplained definition.
Continue with study guide U7 and exercise sheet U7.

## 7.2 Lecture 14 · Intrinsic trace and determinant

**Learning goals and prerequisites**
Define trace without selecting a basis, define determinant through the top exterior power, and derive invariance and multiplicativity.

**Recall and approach.** Recall a finite projective module has a finite dual basis even when it has no ordinary basis.

For a finite free module of rank $n$, $\wedge^n M$ is a free rank-one module. Every endomorphism of it is multiplication by a unique scalar, independent of a chosen generator.

**Definition 17 · Determinant**
For finite free $M$ of rank $n$, define $\det(T)$ by $\wedge^n T = \det(T) 1_{\wedge^n M}$. In a basis this is exactly the coefficient of $e_1 \wedge \dots \wedge e_n$ in $Te_1 \wedge \dots \wedge Te_n$.

**Proof strategy.** Define determinant on the top exterior power; for trace, expand one dual basis in another and interchange finite sums.

**Theorem 30 · Determinant laws**
The determinant is independent of basis, $\det(TS) = \det(T)\det(S)$, and $\det(1) = 1$.

**Proof.** The definition is made on the intrinsic module $\wedge^n M$, so changing a basis only changes a generator of this rank-one module by a unit and does not change the scalar of its endomorphism. Functoriality gives $\wedge^{n(TS)} = (\wedge^n T)(\wedge^n S)$, hence multiplication of their scalars. The identity law follows the same way. Expanding the wedge in a basis gives the usual signed-permutation formula. □

To extend trace beyond free modules, let $M$ be finite projective. Since it is a summand of a finite free module, there are elements $m_i \in M$ and linear functionals $\varphi_i : M \to R$ with $m = \sum_i \varphi_{i(m)} m_i$ for every $m$. This is a dual basis, not necessarily an ordinary basis: take standard vectors in the containing free module, project them to $M$, and restrict the coordinate functionals to $M$.

**Definition 18 · Trace on a finite projective module**
Define $\operatorname{tr}(T) = \sum_i \varphi_{i(Tm_i)}$ using any finite dual basis.

**Theorem 31 · Independence and cyclicity**
This trace is independent of the dual basis and satisfies $\operatorname{tr}(TS) = \operatorname{tr}(ST)$.

**Proof.** For another dual basis $n_j, \psi_j$, expand $m_i = \sum_j \psi_{j(m_i)} n_j$. Then $\sum_i \varphi_{i(Tm_i)} = \sum_{i,j} \psi_{j(m_i)} \varphi_{i(Tn_j)} = \sum_j \psi_{j(Tn_j)}$, using the first dual-basis identity on $Tn_j$. This proves independence. For cyclicity expand $Sm_i$ in the dual basis to get $\operatorname{tr}(TS) = \sum_{i,j} \varphi_{j(Sm_i)} \varphi_{i(Tm_j)}$. Commutativity of $R$ and interchanging $i, j$ give the corresponding formula for $\operatorname{tr}(ST)$. □

For a finite locally free module of constant rank $n$, the determinant line is $\det(M) = \wedge^n M$, a locally free rank-one module, and the intrinsic determinant of $T$ is the induced endomorphism of this line. On every localization where $M$ has a basis it is described by the usual determinant scalar, and changing that basis does not alter the scalar. This formulation remains meaningful when a global basis does not exist.

**Example 25 · An operator and a cyclic representation**
A representation of a cyclic group of order $n$ on $V$ is a linear operator $T$ satisfying $T^n = 1$. It is a module over $K[t]/(t^n - 1)$. Trace and determinant of $T$ are independent of the chosen coordinate description, and $\det(T)^n = 1$.

**Example 26 · Trace as a rank-one contraction**
For $T(m) = \varphi(m)v$ and dual basis $(m_i, \psi_i)$, the trace is $\sum_i \varphi(m_i)\psi_{i(v)}$. Since $v = \sum_i \psi_{i(v)} m_i$, linearity of $\varphi$ makes this sum $\varphi(v)$. No global basis of $M$ was required.

This formula is a useful consistency test for an intrinsic definition: in coordinates it becomes the familiar sum of diagonal entries of an outer product. The dual-basis proof explains why the same scalar is obtained after changing the finite family used to reconstruct elements. Independence is proved, not assumed because the expression looks coordinate-like.

**Checkpoint 7.2.** Why does replacing a matrix by $P^{-1}AP$ preserve trace and determinant?

**Takeaways.** Trace contracts an endomorphism against a dual basis; determinant measures its action on a top exterior power. Both are structural, not coordinate-dependent recipes.
Continue with study guide U7 and exercise sheet U7.

# 8 Unit 8 · Noetherian methods and applications

## 8.1 Lecture 15 · Finiteness conditions and Hilbert's basis theorem

**Learning goals and prerequisites**
Prove equivalent Noetherian conditions, transfer them through exact sequences, and prove Hilbert's basis theorem.

**Recall and approach.** Recall a submodule of a finite free module over a Noetherian ring is finitely generated.

**Definition 19 · Noetherian module**
A module is Noetherian if every increasing chain of submodules stabilizes. A commutative ring is Noetherian if it is Noetherian as a module over itself, so the chains in question are chains of ideals.

**Proof strategy.** Replace a chain condition by finite generation, then lower polynomial degree using finitely many leading coefficients.

**Theorem 32 · Equivalent finiteness condition**
A module is Noetherian exactly when every submodule is finitely generated.

**Proof.** If every submodule is finitely generated, the union of an increasing chain is generated by finitely many elements, all lying in one chain member; the chain stabilizes there. Conversely, if a submodule were not finitely generated, successively choose an element outside the submodule generated by all previous choices. This produces a strictly increasing chain, contradicting the ascending chain condition. □

**Theorem 33 · Exact-sequence criterion**
In $0 \to A \to B \to C \to 0$, the module $B$ is Noetherian exactly when both $A, C$ are Noetherian. Consequently every finitely generated module over a Noetherian ring is Noetherian.

**Proof.** Submodules of a Noetherian module are Noetherian, and submodule correspondence proves the same for quotients. Conversely for a submodule $N$ of $B$, the intersection with $A$ and its image in $C$ are finitely generated. Lift a finite generating set of the image to $N$ and combine it with generators of the intersection. Subtracting the lifted image expression from any element of $N$ leaves an element of the intersection, so these finitely many elements generate $N$. Finite direct sums of $R$ are therefore Noetherian by induction, and every finitely generated module is their quotient. □

**Theorem 34 · Hilbert's basis theorem**
If $R$ is Noetherian, then $R[x]$ is Noetherian. Hence every polynomial ring in finitely many variables over $R$ is Noetherian.

**Proof strategy.** Choose finitely many leading coefficients, reduce high degrees, and handle the bounded-degree remainder as a submodule of a finite free module. **Proof.** Let $I$ be an ideal of $R[x]$. The set of leading coefficients of its nonzero polynomials, together with zero, is an ideal: align degrees by multiplying by powers of $x$ before adding, and scale coefficients by ring elements. Choose generators $a_i$ of this coefficient ideal and corresponding $f_i \in I$ of degrees $d_i$. If $I = 0$, there is nothing to prove. Otherwise put $N = \max_i d_i$. The polynomials of degree less than $N$ form a finite free $R$-module, so its intersection with $I$ is generated as an $R$-module by finitely many $g_j$. For $f \in I$ of degree $d \geq N$, express its leading coefficient as $\sum_i r_i a_i$ and subtract $\sum_i r_i x^{d-d_i} f_i$. The degree strictly decreases. Induction reduces to

degree less than $N$, where the $g_j$ suffice. Thus the $f_i, g_j$ generate $I$ as an ideal. Induct on the number of variables. □

**Example 27 · Why the bounded-degree remainder cannot be skipped**
Choosing polynomials whose leading coefficients generate the coefficient ideal lets us reduce any sufficiently high degree. It does not necessarily express a low-degree remainder using those same polynomials, because multiplying by a negative power of $x$ is not allowed. The proof therefore collects the ideal's degree-less-than-$N$ part as a submodule of $R^N$ and supplies separate finite generators for it.

This is the logical purpose of the two finite lists in Hilbert's proof. A degree-lowering algorithm without a base case is not a completed induction. When $N = 0$, the bounded-degree part is zero and the argument already ends at the zero polynomial.

**Checkpoint 8.1.** Why does the theorem not directly apply to a polynomial ring in infinitely many variables?

**Takeaways.** Noetherianity makes subobjects finitely describable. Hilbert's theorem says adjoining finitely many polynomial variables preserves that control.
Continue with study guide U8 and exercise sheet U8.

## 8.2 Lecture 16 · Algebraic structure in geometry and representations

**Learning goals and prerequisites**
Use quotients, localization, tensors, and finiteness together in concrete applications.

**Recall and approach.** Recall exactness of localization, right exactness of tensors, and distinct-root interpolation.

For polynomial equations over a field, the coordinate ring records algebraic relations among coordinate functions. We use this as a motivating algebraic construction; a full equivalence between varieties and rings requires additional hypotheses and belongs to a later algebraic-geometry course.

**Example 28 · Two intersecting axes**
Let $A = K[x, y]/(xy)$. The residue classes of $x, y$ are nonzero but their product is zero. Localizing at powers of $x$ makes $x$ a unit and forces $y = 0$, so $A_x \cong K[x, x^{-1}]$. This describes the part of the horizontal axis away from the intersection. The formula defines inverse homomorphisms: send $y$ to zero and retain $x$ and its inverse.

**Proof strategy.** Express relations through quotients, invert only declared scalars, and use interpolation idempotents to split an operator.

**Theorem 35 · Finite presentations over Noetherian rings**
Every finitely generated module over a Noetherian ring is finitely presented.

**Proof.** Choose a surjection $R^n \to M$. Its kernel is a submodule of the Noetherian module $R^n$, hence is generated by finitely many elements. Mapping a finite free module onto that kernel gives a presentation $R^m \to R^n \to M \to 0$. □

In a computation, this means finitely many generators have finitely many generating relations, even though there are infinitely many consequences of those relations. It does not mean every algorithm will be fast, nor that a specific chosen relation list is already complete.

**Example 29 · Specializing a module**
For an $R$-module $M$ and maximal ideal $\mathfrak{m}$, the tensor $M \otimes_R R/\mathfrak{m}$ is $M/\mathfrak{m}M$, a vector space over the residue field. A matrix presentation can therefore be specialized by reducing each entry modulo $\mathfrak{m}$. Right exactness ensures the resulting cokernel is the specialized module.

**Theorem 36 · Diagonalization of a finite-order operator**
Let $T^n = 1$ on a finite-dimensional $K$-space. If $t^n - 1$ splits into distinct linear factors over $K$, then the space is the direct sum of the eigenspaces of $T$.

**Proof.** Let its distinct roots be $\lambda_1, \ldots, \lambda_n$. Lagrange interpolation gives polynomials $e_i$ with $e_{i(\lambda_j)} = \delta_{ij}$. Modulo $t^n - 1$, they satisfy $\sum_i e_i = 1$, $e_i e_j = 0$ for $i \neq j$, and $(t - \lambda_i)e_i = 0$. Substitute $T$ to obtain mutually orthogonal projections summing to the identity, with images in the corresponding eigenspaces. Conversely $e_{i(T)}$ acts as identity on its own eigenspace and zero on the others. These projections give the direct-sum decomposition. □

The distinct-root assumption holds when the characteristic does not divide $n$ and all $n$th roots of unity lie in $K$. In characteristic dividing $n$, nontrivial nilpotent behavior can occur instead. This links the field-theory derivative criterion to the module theory of a group action.

**Example 30 · Characteristic changes an operator decomposition**
Over a field of characteristic two, let $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $T = 1 + N$. Then $N^2 = 0$ and $T^2 = 1 + 2N + N^2 = 1$, but $T$ is not the identity and is not diagonalizable: its only eigenvalue is one, so a diagonalizable such matrix would be the identity. The polynomial $t^2 - 1 = (t - 1)^2$ has a repeated root.

This checks the exact hypothesis in the interpolation proof. Distinct roots make denominators $\lambda_i - \lambda_j$ invertible and supply separate projections; repeated roots remove that construction and allow nilpotent parts. The field theory and module theory are describing the same obstruction.

**Checkpoint 8.2.** What does tensoring a presentation with a residue field preserve, and what may it lose?

**Takeaways.** The same universal constructions organize field adjunctions, module presentations, local calculations, and operator decompositions. Use the final assessments to move deliberately between these viewpoints.
Continue with study guide U8 and exercise sheet U8.